

Experiment No. 9

Aim: Design and Verification of Full Adder using logic gates.

Requirement: Digital IC Trainer Kit, Power Supply and Connecting Wires.

Procedure/ Design Steps:

1. Implement the truth table for full adder
2. Simplify the output using K-map
3. Implement the circuit using trainer kit
4. Verify the truth table of all the outputs (sum and carry)

Theory:

An adder is a digital logic circuit in electronics that implements addition of numbers. In many computers and other types of processors, adders are used to calculate addresses, similar operations and table indices in the ALU and also in other parts of the processors. These can be built for many numerical representations like excess-3 or binary coded decimal. Adders are classified into two types: half adder and full adder. The half adder circuit has two inputs: A and B, which add two input digits and generate a carry and sum. The full adder circuit has three inputs: A and C, which add the three input numbers and generate a carry and sum. This article gives brief information about half adder and full adder in tabular forms and circuit diagrams.

Full Adder Circuit

An adder is a digital circuit that performs addition of numbers. The full adder adds 3 one bit numbers, where two can be referred to as operands and one can be referred to as bitcarried in. And produces 2-bit output, and these can be referred to as output carry and sum.

Truth table for full adder is shown in Table 13.

The circuit implementation for full adder is shown in Figures 13

$$\begin{aligned} S &= \overline{A} \overline{B} C_{in} + \overline{A} B \overline{C}_{in} + A \overline{B} \overline{C}_{in} + A B C_{in} \\ &= \overline{A} (\overline{B} C_{in} + B \overline{C}_{in}) + A (\overline{B} \overline{C}_{in} + B C_{in}) \\ &= \overline{A} (B \oplus C_{in}) + A (\overline{B} \oplus \overline{C}_{in}) \end{aligned}$$

$$S = A \oplus B \oplus C_{in}$$

Table: Truth Table for Full Adder

Inputs			Outputs	
A	B	CIN	COU T	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Diagram:

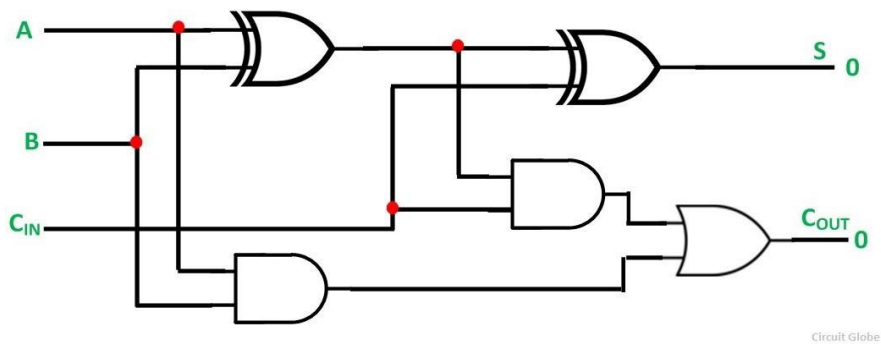


Figure 13: Full adder

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Viva - Voce Questions 6.1

Question 6.1 *Design 2 bit adder using full adder*